// JSON stands for JavaScript Object Notation // it is a way of representing data that is widely // used on the web for sharing and loading data // JSON data is structured as objects, looks just // like regular JS objects, except the keys in JSON are // strings.. unlike a true JS object, JSON is a string // so, when you load JSON into a JS script, job 1 is // to parse or convert the JSON from string to actual // JS object

// dataset as an array of objects const antiques = [{ itemID: 1, picName: 'tang0.jpg', dynasty: 'Tang', category: 'bronze', item: 'statue', sold: true, price: 1000 },

{ itemID: 2, picName: 'song0.jpg', dynasty: 'Song', category: 'ceramic', item: 'bowl', sold: false, price: 1500 },

{ itemID: 3, picName: 'ming0.jpg', dynasty: 'Ming', category: 'wood', item: 'chair', sold: true, price: 2200 }];

// print the following using object.property syntax // console.log(antiques[0].item); // statue // console.log(antiques[2].category); // wood // console.log(antiques[0].dynasty); // Tang // console.log(antiques[1].category); // ceramic // console.log(antiques[2].item); // chair

// in JSON all the keys are in double quotes -- JSON values only get quotes if they are strings

const myJson = '{"result":true, "count":42}'; const myObj = JSON.parse(myJson); console.log(myObj.count); // expected output: 42 console.log(myObj.result); // expected output: true

const myAntique = '{"itemID":1, "picName":"tang0.jpg", "dynasty":"Song", "category":"ceramic", "item": "bowl", "sold": false, "sold": 1500}'; console.log('myAntique:', myAntique); // this is a JSON string console.log('myAntique.picName:', myAntique.picName); // as a string, it has no properties to access

// parse the JSON string into an object with JSON.parse() const myAntiqueObj = JSON.parse(myAntique); // parse means to break into little pieces and separate console.log('myAntiqueObj:', myAntiqueObj); // now this is an actual object with properties console.log('myAntiqueObj.picName:', myAntiqueObj.picName); // we can now access the properties console.log(myAntiqueObj.dynasty); console.log(myAntiqueObj.category); console.log(myAntiqueObj.item); console.log(myAntiqueObj.sold); // put the entire json string in backticks, so that you can hit enter and tab to make it more readable const myJSONArr = `[{ "itemID":1, "picName":"tang0.jpg", "dynasty":"Tang", "category":"ceramic", "item": "camel", "sold":false, "sold": 1500 },

{ "itemID":2, "picName":"song0.jpg", "dynasty":"Song", "category":"ceramic", "item":"bowl", "sold":true, "sold":2500 },

{ "itemID":3, "picName":"ming0.jpg", "dynasty":"Ming", "category":"wood", "item":"chair", "sold":true, "sold":7500 }]`;

console.log('myJSONArr:', myJSONArr); console.log(myJSONArr[0].dynasty); console.log(myJSONArr[1].item); console.log(myJSONArr[2].category);

const objArr = JSON.parse(myJSONArr); console.log('objArr:', objArr); console.log(objArr[0].dynasty); console.log(objArr[1].item); console.log(objArr[2].category);

// loadJSON method to open the JSON file. function loadJSON(path, success, error) { var xhr = new XMLHttpRequest(); xhr.onreadystatechange = function () { if (xhr.readyState === 4) { if (xhr.status === 200) { success(JSON.parse(xhr.responseText)); } else { error(xhr); } } }; xhr.open('GET', path, true); xhr.send(); }

loadJSON("https://jsonplaceholder.typicode.com/posts", myData,'jsonp');

```
function myData(Data) {

// Output only the details on the first post console.log(Data[0]);

// output the details of first three posts console.log("First three posts"); for(var i=0; i<3; i=i+1) {
console.log(Data[i]); } // output the id field of first five elements. console.log("First five ID"); for(var i=0; i<5;
i=i+1) { console.log(Data[i].id); } }
```